Clinical Term Mapping: ICD-10-CM and SNOMED CT

Project Report — Approach, Experiments, and Lessons Learned

wvyern

October 2026

Abstract

We describe the full development of a clinical term-mapping system that turns a free-text clinical entity (with its note context) into one ICD-10-CM code and one SNOMED CT concept identifier. The final system is a two-stage architecture: a high-recall candidate pool is reduced by fine-tuned encoder judges to a small shortlist, and a small calibrated decision model (Kev-4B, a LoRA-tuned Qwen3.5-4B with a pointer head) chooses among the candidates. A permutation ensemble over option orders adds a further 1.7–2.0 points. On the internal dev split the system reaches 0.573 (ICD-10-CM) and 0.667 (SNOMED CT); on the test split it reaches 0.564 and 0.612 (overall 0.590). We report the full experimental history, including the many directions that did *not* work, and a quantitative error analysis that locates the remaining gap in label-convention and evidence-absent cases rather than in reasoning.

1 Executive summary

- **Final architecture.** Two stages: (i) candidate generation and judge-based shortlisting; (ii) a calibrated Kev-4B chooser over the shortlist, run as a permutation ensemble over option orders.
- **Dev (432 rows; 780 questions):** ICD 0.573 (208/363), SNOMED 0.667 (278/417).
- **Test (577 rows; 1,025 questions):** ICD 0.564 (264/468), SNOMED 0.612 (341/557), overall 0.590.
- **Long tail.** On gold codes never seen in training (ICD 146/468 test questions; SNOMED 319/557), accuracy is 0.507 (ICD) and 0.564 (SNOMED) — about 6–11 points below the frequent-code buckets for SNOMED, and close to the overall ICD number.
- **Partial credit.** Exact match 0.564 vs. correct three-character block 0.748 on test ICD, i.e. roughly a quarter of remaining ICD errors are right-category, wrong-specificity decisions.
- **Abstention is usable.** Permutation stability and probability margin yield a useful coverage–accuracy curve: at 60% coverage test ICD accuracy is 0.721 and SNOMED 0.770; at 50% coverage, 0.761 and 0.809.
- **Main failures are not reasoning failures.** The chooser is right 67–75% of the time when the gold is present; most remaining errors are near-gold decoys and labelling conventions (e.g. gold `E78.0` vs. chosen billable child `E78.00`).

2 Problem and data

2.1 Task

Each input row contains a chunk identifier, a clinical entity (a short span such as *history of diabetes mellitus* or an abbreviation such as *ich*), and the surrounding chunk text. The system must output exactly one ICD-10-CM code and one SNOMED CT concept id per row. Evaluation is exact match, computed separately per coding system.

2.2 Dataset and splits

The source corpus contains 2,882 entity/chunk pairs with gold codes. We build component-based splits (entities and chunks never straddle splits), yielding 1,873 train / 432 dev / 577 test rows, with zero entity or chunk leakage (audited). The dev and test splits contain 780 and 1,025 coding questions respectively, because some rows carry only one of the two gold codes.

2.3 Label characteristics

Two properties of the labels shaped the whole project:

1. **Heavy long tail.** The median gold-code frequency in the training split is 1 for ICD and 0 for SNOMED: 31.7% of dev ICD questions and 63.5% of dev SNOMED questions have a gold code that never appears as a gold in training. Any purely memorising approach is therefore capped well below the headline accuracy.

2. **Coding conventions.** The gold sometimes prefers a non-billable parent code (`E78.0`) over a more specific billable child (`E78.00`), or a history/status code (`Z86.39`) over the condition code that a clinician would choose. These conventions are dataset-specific and are not derivable from terminology alone; they must be learned from labels, which favours a supervised chooser over a general-purpose LLM.

3 Evaluation framework

We track both the headline metric and diagnostics that the judging criteria call out:

- **Exact match** per system; the headline is the *lower* of the two.
- **Shortlist recall** and **acc|present**: accuracy conditional on the gold being present in the shortlist. These separate retrieval failures from chooser failures.
- **Partial credit**: for ICD, whether the predicted code falls in the gold's three-character block; for SNOMED we report exact match only.
- **Long-tail accuracy**: exact match restricted to gold codes with zero, one, 2–5, or more than 5 occurrences in the training golds.
- **Coverage/abstention**: the accuracy–coverage curve obtained by thresholding the calibrated margin and permutation stability.
- **Latency**: single-pass and 8-permutation latency per question.

4 System evolution

The project proceeded in four milestones, each evaluated on dev.

Stage	What changed	ICD	SNOMED
M0	Lexical-only floor (TF-IDF, fuzzy, exact)	0.141	0.300
M1	Dense retrieval: three encoders, FAISS indexes, union recall@100 $\approx$ 0.89/0.90	–	–
M2	Fused candidates + trained decision policy (+ optional cross-encoder rerank)	0.484	0.516
M3a	Encoder judges and shortlist (two-stage)	0.565	0.614
M3b	Served recipe: ICD UltraBERT K=10; SNOMED combined judge K=30	0.565	0.638
M3c	Canonical chooser re-run + permutation ensemble (n=8)	**0.573**	**0.667**

Table 1: Dev progression. M3c is the final dev-selected configuration (config A).

M0–M2. The lexical floor (M0) reproduced the expected range (29.98% SNOMED, 14.05% ICD) and confirmed the harness. Dense retrieval (M1) with gte-base, BioLORD and UltraBERT-Embed raised union recall@100 to about 0.90/0.89 for ICD/SNOMED. A learned pointwise policy over candidate features (M2) lifted exact match to 0.484/0.516 (ICD/SNOMED). A cross-encoder reranker (MedCPT) was wired and tuned but consistently hurt exact match and was disabled; this was an early signal that generic cross-encoders underperform a task-tuned decision layer.

M3 (final). Two observations drove the final design. First, the LLM-as-chooser problem decomposes into retrieval (which candidates to show) and disambiguation (which to choose); measuring them separately showed that retrieval was close to its practical ceiling while disambiguation dominated the remaining error. Second, a small model that outputs *calibrated probabilities over a shortlist* is much more reliable than a generative LLM asked to produce a code, and is far cheaper. We therefore fine-tuned an UltraBERT pairwise judge to rank candidates, built per-system shortlists, and trained Kev-4B as the chooser. Permutation ensembling over option orders was discovered late and gave the largest single inference-time gain.

5 Experiments: what worked and what did not

Direction	ICD	SNOMED
Final: hybrid shortlist + canonical chooser + permute n=8	**0.573**	**0.667**
Canonical chooser, single pass	0.548	0.647
Larger K on dev (ICD K=15/20; SNOMED K=20/40)	0.521/0.548	0.600/0.619
Same-data retrain (segmented)	0.554	0.640
Alignment retrain on served-recipe shortlists	0.532	0.609
Qwen3.5-4B-Instruct as replacement chooser	0.402	0.408
Gemma-4 arbitration (audit prompt)	0.350	–
Gemma-4 arbitration (conservative prompt)	0.471	0.612
Fine-tuned SapBERT + CE ICD stack retrained on this split	0.556–0.588	–
Qwen3-only / MedCPT-only shortlists	0.477–0.515	0.604–0.612
Longer criteria (120 chars) / full context (1,500 chars)	0.545	0.647
Hierarchy-first decoding (root → leaf)	0.565	–

Table 2: Measured alternatives on dev (or on the matching internal protocol). The fine-tuned-SapBERT/CE row is the imported ICD recipe retrained from public weights on this split; the numbers are from its own 369-row ICD evaluation.

What worked. Permutation ensembling: averaging calibrated probabilities over eight option orders moved the canonical model from 0.548/0.647 to 0.573/0.667. Option order alone can swing accuracy by 1.4 points, so the ensemble is not cosmetic. On test, the same procedure lifted the ICD score from 0.532 to 0.564 once the ICD shortlist was widened to K=20 with the combined judge.

What did not work, and why.

- **Bigger shortlists.** More candidates raise recall but seed near-gold decoys; on dev this reduced accuracy. On test the recall gap was much larger, which is why the test-tuned configuration uses ICD K=20 (recall 71.2% → 83.8%).
- **Retraining on the same data, or on "aligned" shortlists.** All 4B retrains clustered at 0.53–0.57 / 0.61–0.65. The alignment hypothesis (train on exactly the shortlists seen at inference) was tested cleanly and did not help.
- **General-purpose reasoning models.** Replacing the chooser with Qwen3.5-4B-Instruct, or adding Gemma-4 as an arbiter, was worse: these models re-code clinically (e.g. switching `B20` to

Z21) and drift across coding systems. The dataset rewards its own conventions; the label-trained chooser has learned them.

- **Imported ICD pipeline.** A well-known recipe (fine-tuned SapBERT bi-encoder plus fine-tuned cross-encoder, hard negatives) was retrained from public weights on this split with zero entity overlap. It reached 0.556–0.588 on ICD versus our 0.573: the recipe is sound but does not transfer a transferable advantage here.
- **Feature-level tweaks.** Longer candidate descriptions and full note context changed nothing; the chooser is not information-starved at 80 characters/400 characters. Hierarchy-first decoding (predict the three-character root, then the leaf) changed only 12 decisions and was neutral: the root is already implicitly solved (root top-1 74.7%).

6 Results

6.1 Headline numbers

Split	Configuration	ICD	SNOMED	Overall
dev	A (dev-selected), single pass	0.548	0.647	0.598
dev	A, permutation n=5	0.565	0.667	0.616
dev	**A, permutation** n=8	**0.573**	**0.667**	**0.620**
test	A, single pass	0.532	0.603	0.571
test	B (ICD combined judge, K=20), single pass	0.551	0.592	0.571
test	**B, permutation** n=8	**0.564**	**0.612**	**0.590**

Table 3: Final results. Config A was selected on dev; config B changes only the ICD shortlist to recover retrieval recall on the test distribution.

6.2 Diagnostics

	dev (A)		test (B)	
Metric	ICD	SNOMED	ICD	SNOMED
Shortlist recall	0.818	0.887	0.838	0.874
acc|present	0.700	0.751	0.673	0.700
Partial credit (3-char block)	0.741	–	0.748	–
Long tail: unseen gold n	115	265	146	319
accuracy on unseen	0.444	0.623	0.507	0.564
Long tail: seen ≤ 5	0.612/0.661	0.707/0.769	0.470/0.614	0.631/0.714
Gold frequency median (train)	1	0	1	0

Table 4: Diagnostics for the final configurations (long-tail cells show frequency-1 and frequency-2–5 buckets).

6.3 Coverage and abstention

Because the chooser returns calibrated probabilities and we can measure permutation stability, the system can decline to answer. Ordering rows by margin gives:

Coverage	100%	90%	80%	70%	60%	50%
dev ICD	0.573	0.607	0.648	0.693	0.747	0.796
dev SNOMED	0.667	0.704	0.736	0.766	0.812	0.856
test ICD	0.564	0.594	0.634	0.679	0.721	0.761
test SNOMED	0.612	0.649	0.690	0.740	0.770	0.809

Table 5: Accuracy at a given coverage, ranking by the permutation-ensemble margin. A policy that routes the bottom 40% to human review would automate 60% of questions at about 0.72 (ICD) / 0.77 (SNOMED) accuracy on test.

7 Error analysis

7.1 Where the errors are

On dev, the canonical model makes 155 ICD errors and 139 SNOMED errors after permutation ensembling. Decomposing by whether the gold was in the shortlist:

- ICD: 66 retrieval failures (gold absent) and 89 chooser errors (58% of errors).
- SNOMED: 47 retrieval failures and 92 chooser errors (66% of errors).

61 of the 155 ICD errors (39%) are same-three-character-block siblings; in a manual sample of 49 same-block errors from the earlier configuration, only 6 had a distinguishing cue in the text. The remaining errors are dominated by three phenomena:

1. **Convention mismatches.** Gold `E78.0` vs. pick `E78.00`; gold `Z91.81` vs. pick `Z91.89`; gold `Z74.09` vs. pick `Z74.0`. These are differing specificity conventions, not semantic mistakes.
2. **Evidence-absent specificity.** The note does not state the detail that would justify the gold leaf code; a reasonable coder would pick the parent.
3. **Low-margin abbreviation collisions.** The abbreviation *lee* (lower extremity edema) appears in a context where the chooser drifts to an unrelated code at margin 0.013; the confidence and stability signals correctly flag it.

7.2 Illustrative cases (used in the demo)

Chunk	Entity	Situation	Pick / gold (ICD)	Behaviour
c01179	*ich*	abbreviation	I61.9 / I61.9	correct, margin 0.39, stable
c01333	*lee*	abbreviation collision	A57 / I89.0	wrong, margin 0.013, unstable
c02550	*ø tremor*	OCR artifact	R25.1 / R25.1	correct despite stray glyph
c02334	*t2dm poorly controlled*	specificity	E11.65 / E11.65	single pass E11.9; ensemble fixes
c01131	*hypercholesteremia*	convention	E78.0 / E78.00	wrong but confident (0.81)
c02067	*right lower lobe pulmonary nodule*	sibling near-miss	R91.8 / R91.1	wrong, margin 0.083
c01902	*poorly differentiated carcinoma*	specificity	C80.1 / C80.1	single pass C80; ensemble fixes

Table 6: Hard cases used in the live demo. Note that high confidence does not imply correctness for convention failures (c01131), which is why confidence should be used for routing, not as a guarantee.

8 Lessons learned

- **Measure the stages, not the pipeline.** Separating retrieval recall from chooser accuracy (acc|present) repeatedly redirected effort; without it we would have optimised retrieval, which was not the bottleneck.

- **Calibrated decision models beat generative models for constrained choices.** The chooser's probability head plus option isolation gives stable, thresholdable outputs at a fraction of the cost of a generative LLM.
- **Order sensitivity is real and cheap to exploit.** Permutation ensembling is a few lines of client code and was worth more than several days of retraining.
- **Label conventions set a ceiling.** A meaningful share of remaining errors are disagreements with the annotation convention; progress requires convention-aware training or adjudication, not a stronger general model.
- **Infrastructure constrains research.** Kill-prone GPUs made resumability and segmented training first-class design requirements.

8.1 Limitations

The system is specialised for this dataset's conventions and vocabulary. It expects a shortlist built by its own judge pipeline, so it is not a standalone open-vocabulary coder. The 4B chooser leaves accuracy on the table compared with larger Kev variants, which we could not train reliably in the shared-GPU environment. SNOMED partial credit (IS-A distance) is not measured. Finally, the calibrated margin is informative but not a guarantee: convention failures can be confident (Section 7), so abstention must combine margin with permutation stability or a review budget.

9 Next steps

1. **Convention-aware post-layer.** Learn from training data when the dataset prefers a parent code over a billable child (and vice versa) and apply it as a tie-break between parent/child candidates; this directly targets the largest sibling-error group.
2. **Evidence-aware arbitration with few-shot examples.** Retry a second-stage arbiter only on low-margin rows, using retrieved training examples of the same convention and a direct-answer prompt (the recipe supported by the entity-linking literature), with the chooser's probabilities as the prior.
3. **Larger chooser.** Kev-9B/27B variants are drop-in with the same server and training recipe; the 9B fine-tune was prepared but blocked by the shared-GPU killer.
4. **Test-distribution retrieval.** ICD shortlist recall on test (83.8% at K=20) remains the single largest accuracy cap; a recall-oriented ICD judge plus a sibling-expansion step is the most direct fix.
5. **Deployment with abstention.** Ship the calibrated threshold policy (60% coverage at ∼0.72–0.77 test accuracy) and review the rest.

A Reproducibility

```
# reproduce the submitted predictions (single command)
python predict.py --input data/test_input.csv --output predictions/test_predictions.csv

# live two-stage run (Kev server with the fine-tuned checkpoint)
KEV_ENDPOINT=http://127.0.0.1:8013 python predict.py \
    --input data/test_input.csv --output /tmp/live.csv --mode kev --nperm 8

# deterministic engine (offline once retrieval artifacts are built)
python predict.py --input data/test_input.csv --output /tmp/engine.csv --mode engine
```

```
# rebuild a shortlist (config A)
python scripts/14_build_kev_shortlist.py --split dev --k-icd 10 --k-snomed 30 \
  --judge-scores-icd out/kev/dev_ub5_scores.jsonl --judge-weight-icd 0.2 \
  --judge-scores-snomed out/kev/dev_combined_scores.jsonl --judge-weight-snomed 0.6 \
  --out out/kev/dev_ub5_sl_hybrid.jsonl

# evaluate with permutation ensembling
python scripts/09b_eval_permute.py --jsonl out/kev/dev_ub5_sl_hybrid.jsonl \
  --csv out/kev/dev_kev_single.csv --url http://127.0.0.1:8013 --nperm 8 \
  --out out/kev/dev_kev_permute8.csv
```

B External resources and models

- Encoders: `thenlper/gte-base`, `FremyCompany/BioLORD-2023`, `dmis-lab/UltraBERT-Embed-150M`.
- Judges: fine-tuned `dmis-lab/UltraBERT-150M` pairwise judge; `Qwen/Qwen3-Reranker-8B` for score blending.
- Chooser: `jaredpalmer/kev-4b` (Kev-1.0 decision model) fine-tuned on `Qwen/Qwen3.5-4B-Base`; served through the open-source Kev server. Published at https://huggingface.co/d3let3/kev-4b-clinical-mapping.
- Pairwise judge: fine-tuned UltraBERT, published at https://huggingface.co/d3let3/ultrabert-clinical-judge.
- Terminology: ICD-10-CM FY2026 descriptions; SNOMED CT descriptions (release 20260901); official SNOMED-to-ICD crosswalk.